USN | | | | | | | | | | [23TBSCSMA31]

THIRD SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

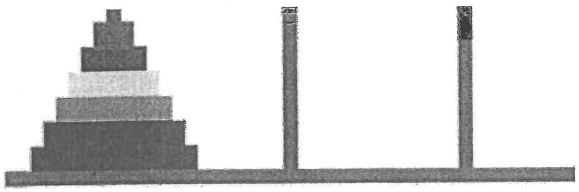
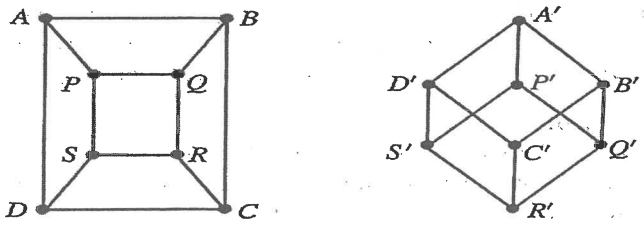
DISCRETE MATHEMATICS AND GRAPH THEORY

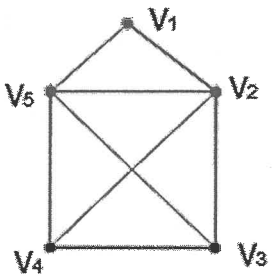
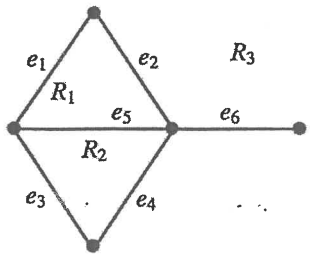
Time: 3 Hours **Maximum Marks: 100**

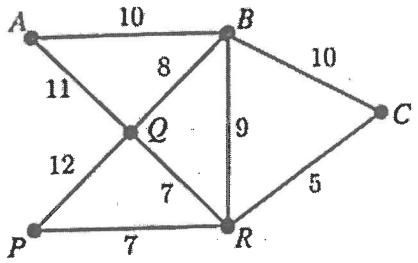
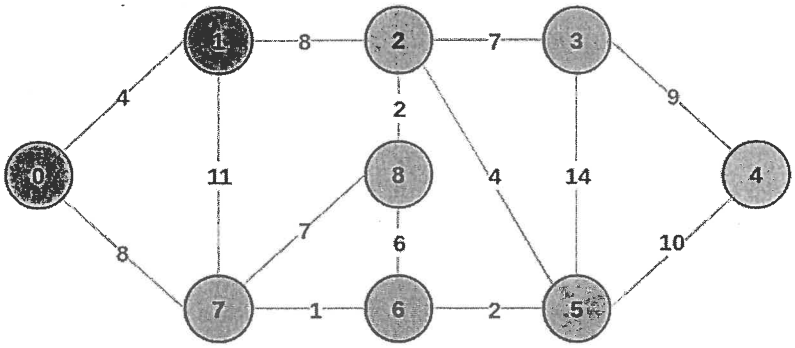
Instructions: **(i) Missing data may be suitably assumed**

(ii) Answer all Questions

		PART-B	5 X 2=10		
1.	**a.**	A license plate consists of two English letters followed by four digits. If repetitions are allowed, how many of the plates have only vowels (A, E, I, O, U) and even digits (0, 2 , 4 , 6, 8)?	L2	CO1	**02**
	b.	Find the sequence generated by the $(3+x)^3$.	L3	CO1	**02**
	c.	Solve the recurrence relation $a_n = 7a_{n-1}$, where $n \geq 1$, given that $a_2 = 98$.	L3	CO1	**02**
	d.	Define Simple graph and Multi graph.	L1	CO1	**02**
	e.	A class room has 25 microcomputers that has to be connected to a wall socket that has four outlets. The connections are made by using extension cords that have four outlets each. What is the least number of cords needed to get computer to use?	L2	CO1	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	How many friends must you have to guarantee that at least 5 of them will have birthdays in the same month?	L3	CO2	**05**
		OR			
	b.	Determine the number of positive integers n such that $1 \leq n \leq 100$ and n is divisible by 2, 3, or 5.	L3	CO2	**05**
3.	**a.**	Determine the coefficient of x^{15} in $\frac{(1+x)^4}{(1-x)^4}$.	L3	CO3	**05**
		OR			
	b.	Using generating function, find the number of nonnegative integer solutions of the equation $x_1 + x_2 + x_3 + x_4 = 25$.	L3	CO3	**05**
4.	**a.**	Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with the initial conditions $a_0 = 1$ and $a_1 = 4$.	L3	CO4	**05**
		OR			
	b.	There are 3 pegs fixed vertically on a table, and n circular disks having holes at their centers and having increasing diameters are slipped onto one of these pegs, with the largest disk at the bottom. The disks are to be transferred, one at a time, onto another peg with the condition that at no time a larger disk is put on a smaller	L3	CO4	**05**

		disk. Determine the number of moves for the transfer of all the n disks, so that at the end the disks are in their original order. 			
5.	**a.**	Prove that in every graph, the number of vertices of odd degrees is even.	L2	CO5	**05**
		OR			
	b.	Verify that the two graphs shown below are isomorphic.  (a) (b)	L3	CO5	**05**
6.	**a.**	Exhibit the following: **i.** A graph which has both Euler circuit and Hamilton cycle **ii.** A graph which has Euler circuit but no Hamilton cycle. **iii.** A graph which has Hamilton cycle but no Euler circuit. **iv.** A graph which has neither Hamilton cycle nor Euler circuit.	L2	CO5	**05**
		OR			
	b.	If a tree T has 3 vertices of degree 2, 2 vertices of degree 3 and 2 vertices of degree 4. Find the number of pendant vertices in T.	L3	CO6	**05**
7.	**a.**	Calculate the number of vertices and the number of leaves in a complete binary tree having 10 internal vertices.	L3	CO6	**05**
		OR			
	b.	Using the merge sorting method sort the following list -1, 0, 2, -2, 3, 6, -3, 5, 1, 4.	L3	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Obtain the rook polynomial for the 3×3 board by using expansion formula.	L3	CO2	**10**
		OR			
	b.	In how many ways can 26 letters of the English alphabets be permuted so that none of the patterns **CAR, DOG, PUN**, or **BYTE** occurs?	L3	CO2	**10**

9.	**a.**	Using generating function, find the number of partitions of n = 6 and write all possible partitions of n=6.	L3	CO3	**10**
		OR			
	b.	Using exponential generating function, find the number of ways in which 4 of the letters in the word **ENGINE** be arranged.	L3	CO3	**10**
10.	**a.**	Solve the recurrence relation $a_n - a_{n-1} - 2a_{n-2} = 4$, given $a_0 = 0$ and $a_1 = -1$.	L3	CO4	**10**
		OR			
	b.	Find the generating function for the recurrence relation. $a_{n+1} - a_n = n^2, \quad n \geq 0$, with $a_0 = 1$. Hence solve it.	L3	CO4	**10**
11.	**a.**	Find the chromatic polynomial for the graph shown below using decomposition theorem. If 5 colors are available, in how many ways can the vertices of this graph be properly colored? 	L3	CO5	**10**
		OR			
	b.	Find the dual of the following planner graph and briefly explain the steps. 	L3	CO5	**10**
12.	**a.**	Explain the following with an example. **i.** Walk **ii.** Trail **iii.** Path **iv.** Circuit **v.** Cycle	L2	CO5	**10**
		OR			

	b.	Using Kruskal's Algorithm, find a minimal spanning tree for the given weighted graph. 	L3	CO6	**10**
13.	**a.**	Obtain an optimal prefix code for the message ROAD IS GOOD. Indicate the code.	L3	CO6	**10**
		OR			
	b.	Using Prim's algorithm, find a minimal spanning tree of the weighted graph. 	L3	CO6	**10**